

Despite common misconceptions, a one-to-one trade-off between underperformance and withdrawal rates does not exist. As the portfolio decreases in size, underperformance impacts a smaller amount of wealth, while real withdrawal amounts for retirement spending do not change. This explains why the 4 percent rule did not become the 3 percent rule in response to the underperformance.

Exhibit 5.12

Maximum Sustainable Withdrawal Rates (MWRs)

Cases: (1) Earn Underlying Index Returns, (2) Underperform Index by 1% Annually

For 50/50 Asset Allocation, 30-Year Retirement Duration, Inflation Adjustments

Using SBBi Data, 1926–2016, S&P 500 and Intermediate-Term Government Bonds

Year	MWR			Year	MWR		
	(Pure)	(1% Less)	Difference		(Pure)	(1% Less)	Difference
1926	7.28	6.47	0.81	1958	5.61	4.9	0.71
1927	7.03	6.24	0.79	1959	4.91	4.29	0.62
1928	6.03	5.35	0.68	1960	4.86	4.25	0.61
1929	5.12	4.54	0.58	1961	4.81	4.21	0.6
1930	5.39	4.79	0.6	1962	4.38	3.83	0.55
1931	5.81	5.16	0.65	1963	4.64	4.07	0.57
1932	7.13	6.34	0.79	1964	4.34	3.82	0.52
1933	6.78	6.02	0.76	1965	4.11	3.62	0.49
1934	5.63	5	0.63	1966	4.03	3.56	0.48
1935	5.79	5.14	0.65	1967	4.41	3.9	0.51
1936	4.91	4.36	0.55	1968	4.18	3.7	0.48
1937	4.35	3.87	0.48	1969	4.19	3.73	0.46
1938	5.55	4.95	0.6	1970	4.83	4.31	0.52
1939	4.75	4.24	0.51	1971	4.81	4.3	0.51
1940	4.8	4.3	0.5	1972	4.64	4.16	0.48
1941	5.19	4.65	0.54	1973	4.44	4	0.44